Drew University - CSCI 260 - Fall 2022
HW7 - C Programming - String and File I/O

HW Info.: Due on 11/26/2022 (T) by 11:59pm, 20 points

Submission: Submit a single C source files *Caesar.c* on Moodle. In the very first line of the program, put your name in the comment with //.

Note: Please follow Drew University's Standard of Academic Integrity and refer to the course syllabus for late submission policy.

Program Description

(20 points) **Caesar Cipher**: Implement a program that reads in a mode (char) and a key (int) from keyboard, and text (char[]) from a file (FILE*) and apply the Caesar cipher to it.

1. At the beginning, the program will ask the user to input a char ['E'/'D'] via the prompt "Please select mode [E/D]: ". (Keep asking for a mode if not entering 'E' or 'D'.)

2. Then, the program will ask the user to input a number (any integer, negative, zero, or positive) via the prompt "Please enter the key: ". (Assume that the input is indeed a number.)

3. Then depending on the mode, in the main function, the program will identify an input file:

   - "plain_in.txt" if the mode was 'E';
   - "cipher_in.txt" if the mode was 'D';

   where the files are located in the same directory as your *main.c*.

4. Then, depending on the mode, the main function will call the corresponding encryption/decryption function, where both of these two functions will take in 2 arguments:

   - the file pointer associated with the input file;
   - the key read from keyboard;

5. Then, in the encryption/decryption function, the input text is processed with the "shifting by key amount" operations, with the following rules:

   - an upper-case letter, after the shifting, stays an upper-case letter (i.e., wrapping around);
   - a lower-case letter, after the shifting, stays a lowercase letter (i.e., wrapping around);
   - otherwise, the ASCII of a char is not shifted at all.
   - Since the key can be of any integer value (e.g. -104 or 4362), the wrapping around should work both ways.

   [**Note** Is there a single line of code that "reduces" the key to its "equivalent" value in terms of the shifting amount?]

6. Lastly, depending on the mode, the shifted text will be written to a file:

   - "cipher_out.txt" if the mode was 'E';
   - "plain_out.txt" if the mode was 'D'.

7. Your program should contain sufficient comments and should at least build without any errors.

8. Submit a single C source files *Caesar.c* on Moodle. In the very first line of the program, put your name in the comment with //.

9. You can use the "sample.txt" file as a test input.